

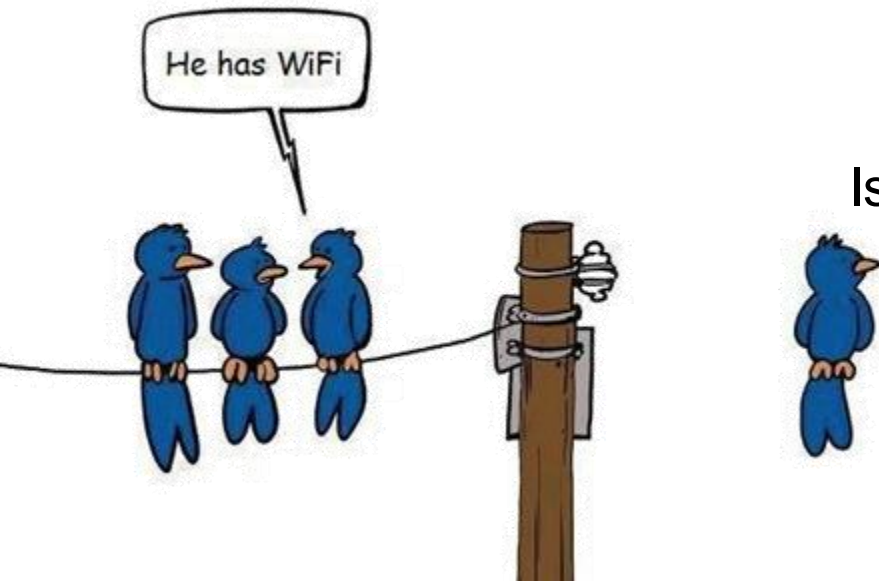
# ECSE 4560/ 6560

## Modern Communication Systems

Lecture 22: Wireless Sensing Part 3:

AOA-TOF

Ish Jain, Fall 2025



# Announcements

- Join Piazza: <https://piazza.com/rpi/fall2025/ecse45606560>
- Per-Class Quiz
  - Posted on LMS; Due Today 11:59 PM
- Attendance Link: <https://forms.office.com/r/p6P9YrN1d8>
  - Today's Magic Word: **AOA-TOF**
- Projects
  - Group formation [MCS\\_Fall25\\_project.xlsx](#)
  - Final Project Presentation and Demos: Dec 9 and Dec 12
  - Project Report and peer evaluation: Finals exam week
- HW/Exams Dates
  - HW6 (MIMO code) due **Mon 11/17**
  - HW7 (Sensing Theory) due **Fri 11/21 – covered in Exam 2**
  - Exam 2: **Friday Nov 21 - Anyone needs makeup exam?**

# Extra Reading on Piazza

- Principles of Modern Wireless Communication Systems Theory and Practice; by Aditya K. Jagannatham, IIT Kanpur Book
  - [https://piazza.com/class\\_profile/get\\_resource/mevvk9kirhg2fu/mgb9esiush64ae](https://piazza.com/class_profile/get_resource/mevvk9kirhg2fu/mgb9esiush64ae)
  - Parts of Chapter 1,2,3,4,7
  - **MIMO Chapter 6**
- Sensing Readings
  - Xiong, Jie, and Kyle Jamieson. "ArrayTrack: A Fine-Grained indoor location system." *NSDI 2013*
    - 2024 SIGMOBILE Test-of-Time Award 

# Today's Class

- **Wireless Sensing**
  - **Localization Techniques**
  - **AOA-TOF based localization**

# Wireless Localization / Positioning

**The process of obtaining a human or object's location using wireless signals**

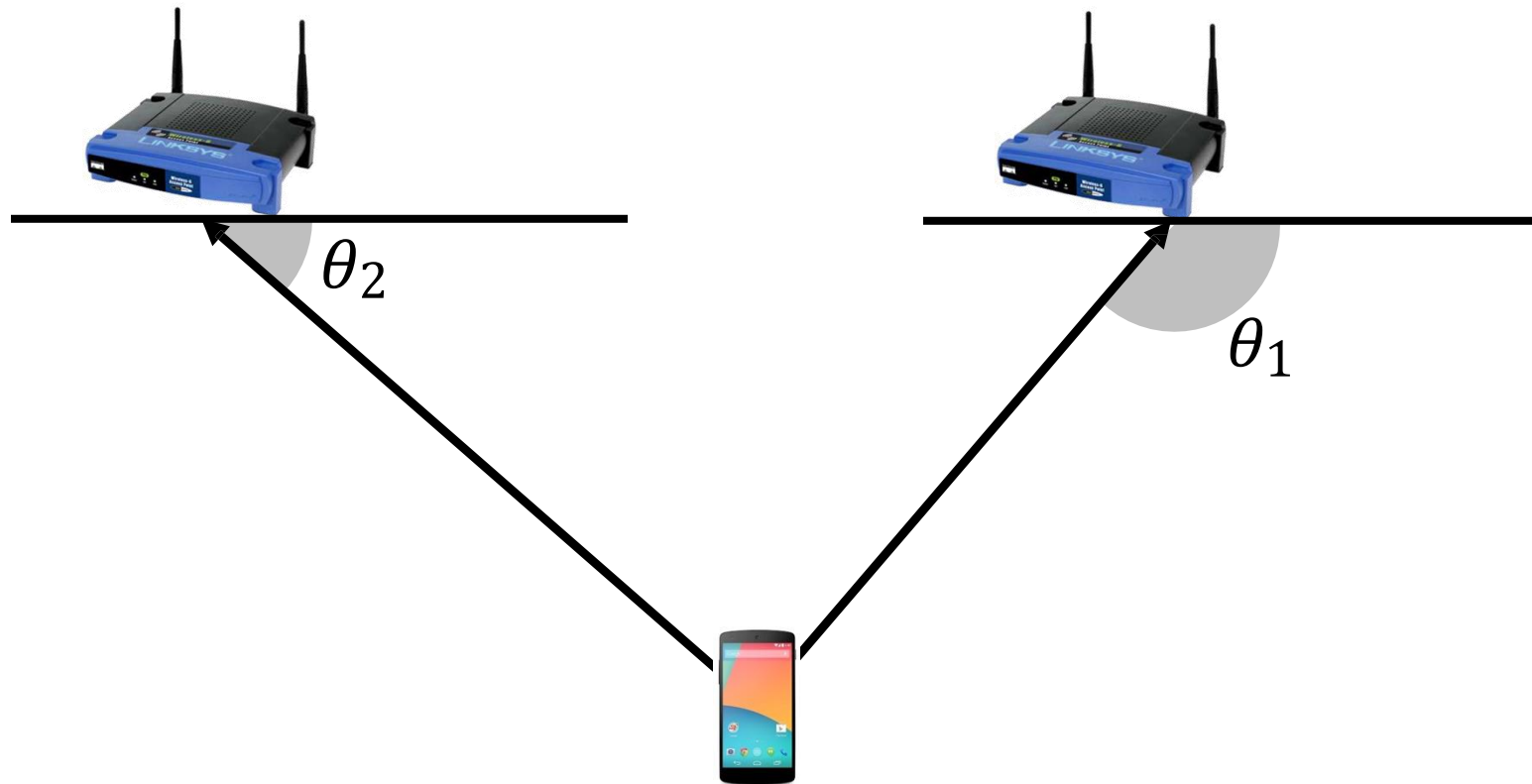
## Applications:

- Navigation: outdoors (GPS) and indoors (e.g., museum)
- Location based services: Tagging, Reminder, Ads
- Virtual Reality and Motion Capture
- Gestures, writing in the air
- Behavioral Analytics (Health, activities, etc.)
- Locating misplaced items (keys)
- Location based security
- Delivery drones



# Method 2: AoA Based Localization

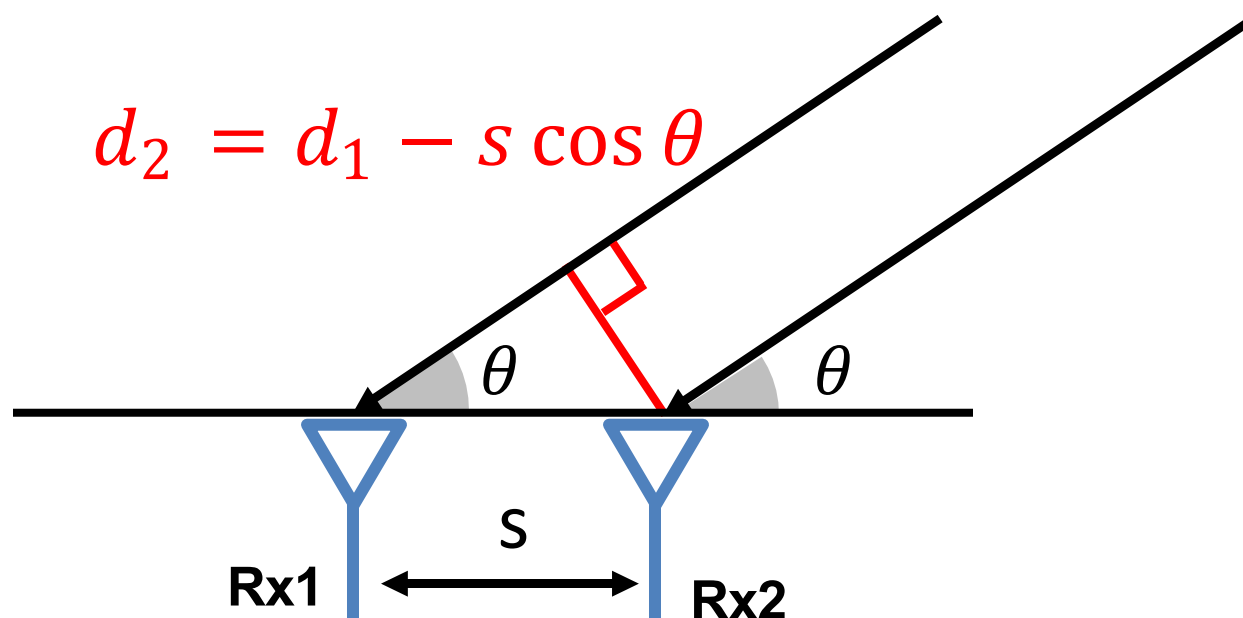
Measure Angle of Arrival (AoA) from device to each AP



**Triangulation**

# Method 2: AoA Based Localization

Measure Angle of Arrival (AoA) from device to each AP

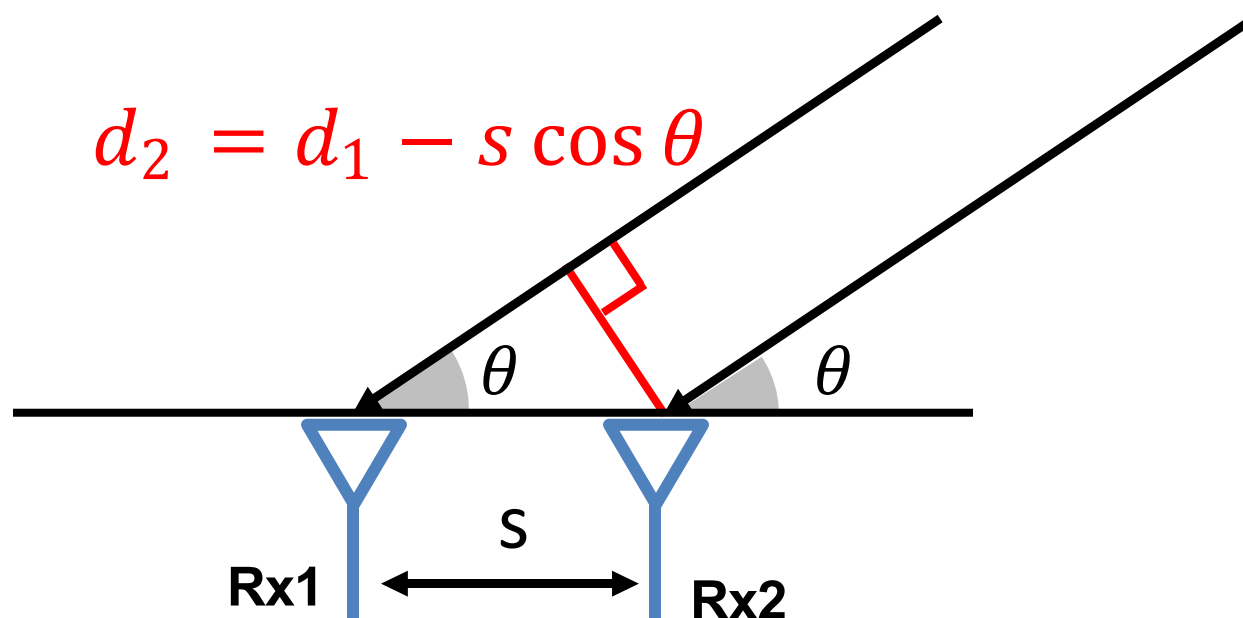


$$h_1 \propto e^{-j2\pi \frac{d_1}{\lambda}} \quad h_2 \propto e^{-j2\pi \frac{d_2}{\lambda}} = e^{-j2\pi \frac{d_1 - s \cos \theta}{\lambda}}$$

$$\Delta\Phi = \angle h_2 - \angle h_1 = 2\pi s \cos \theta / \lambda \bmod 2\pi$$

# Method 2: AoA Based Localization

Measure Angle of Arrival (AoA) from device to each AP



$$\Delta\Phi = \angle h_2 - \angle h_1 = 2\pi s \cos \theta / \lambda \mod 2\pi$$

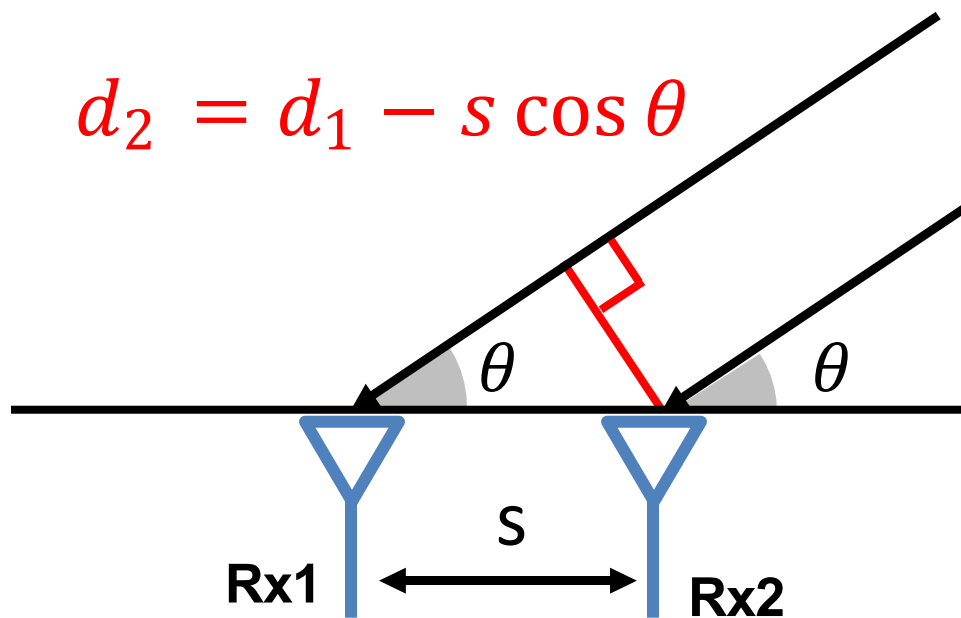
**Ambiguity:**  $\exists \theta_1 \neq \theta_2 \mid \Delta\Phi_1 = \Delta\Phi_2 \mod 2\pi$

**To avoid ambiguity, we want:**  $-\pi \leq \Delta\Phi \leq \pi$



# Method 2: AoA Based Localization

Measure Angle of Arrival (AoA) from device to each AP



We want:

$$-\pi \leq \Delta\Phi \leq \pi$$

We have:

$$-1 \leq \cos \theta \leq 1$$

$$-2\pi \frac{s}{\lambda} \leq 2\pi \frac{s}{\lambda} \cos \theta \leq 2\pi \frac{s}{\lambda}$$

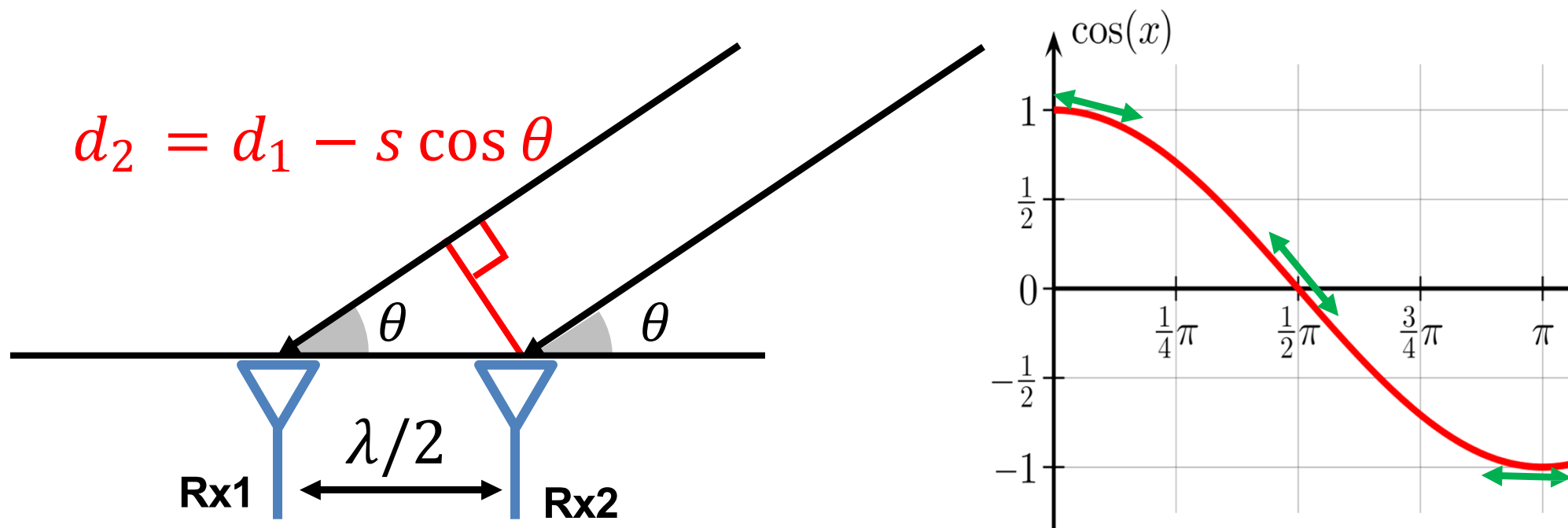
$$-2\pi \frac{s}{\lambda} \leq \Delta\Phi \leq 2\pi \frac{s}{\lambda}$$

Set:  $s = \lambda/2$

$$-\pi \leq \Delta\Phi \leq \pi$$

# Method 2: AoA Based Localization

Measure Angle of Arrival (AoA) from device to each AP



**Pros:** More accurate than RSSI, Simple!

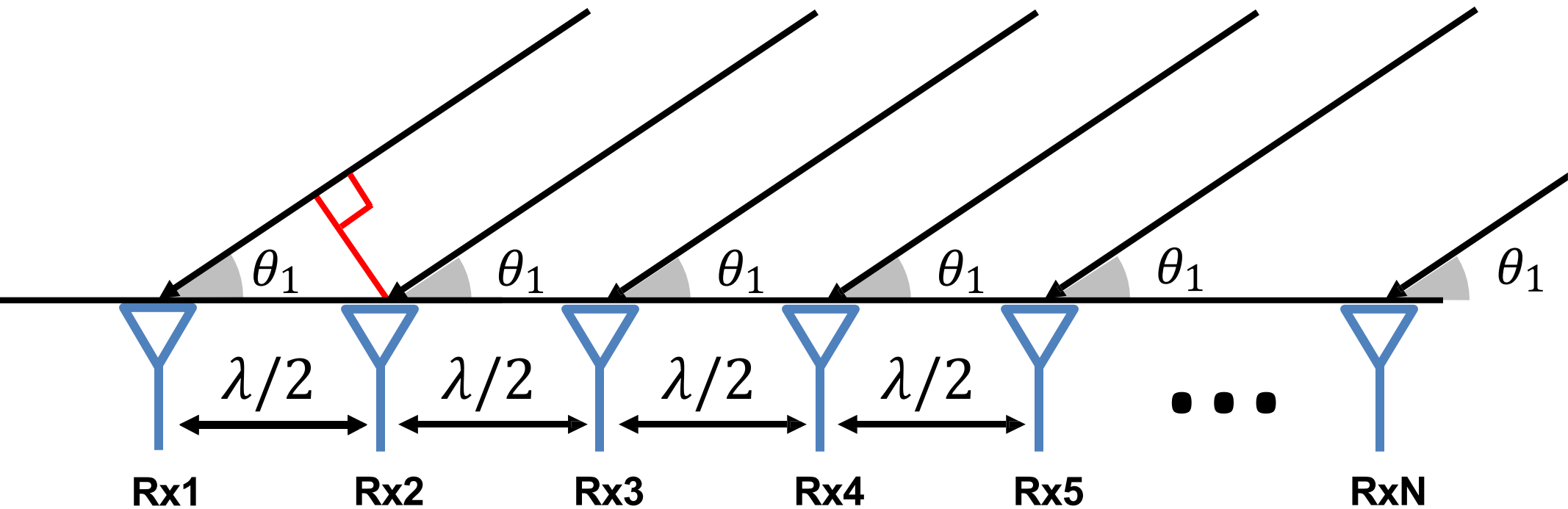
**Cons:** Ambiguity:  $\cos \theta = \cos(-\theta)$

Error not linear with  $\theta$  due to  $\cos \theta$

Requires 2 Antennas separated  $\lambda/2$

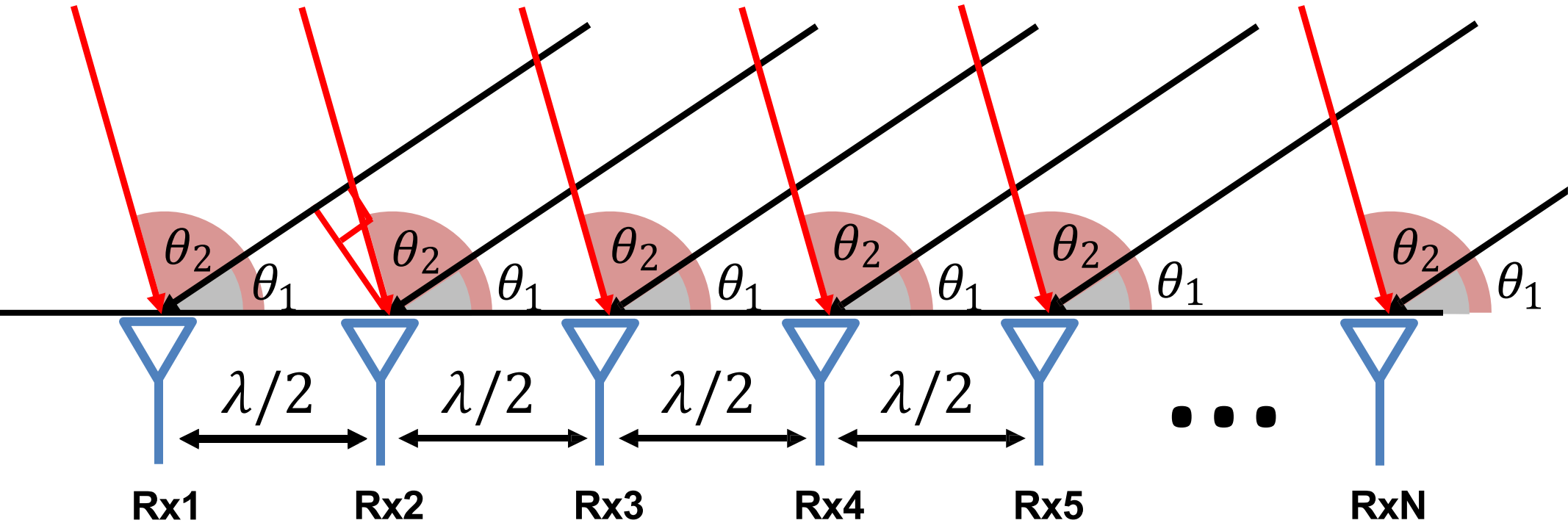
Does not work with multipath!

## Method 3: Antenna Arrays (ArrayTrack Paper)



$$h_k = \alpha_1 e^{-j2\pi \frac{d_1 - k s \cos \theta_1}{\lambda}} = \alpha_1 e^{-j\phi_1 + k\pi \cos \theta_1}$$

# Method 3: Antenna Arrays (ArrayTrack Paper)



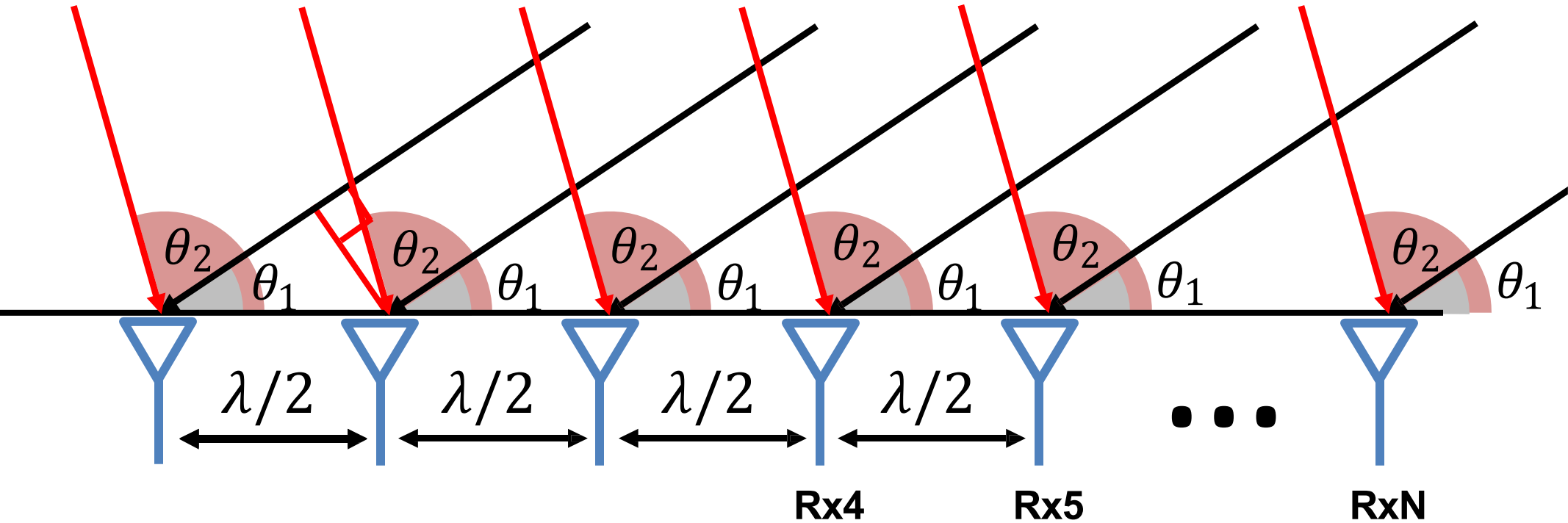
$$h_k = \alpha_1 e^{-j2\pi \frac{d_1 - k s \cos \theta_1}{\lambda}} = \alpha_1 e^{-j\phi_1 + k\pi \cos \theta_1}$$

$$+ \alpha_2 e^{-j2\pi \frac{d_2 - k s \cos \theta_2}{\lambda}} + \alpha_2 e^{-j\phi_2 + k\pi \cos \theta_2}$$

For L paths  $\rightarrow$

$$h_k = \sum_{l=1}^L \alpha_l e^{jk\pi \cos \theta_l - j\phi_l}$$

## Method 3: Antenna Arrays (ArrayTrack Paper)



$$h_k = \sum_{l=1}^L \alpha_l e^{jk\pi \cos \theta_l - j\phi_l}$$

Multipath Profile:  $P(\theta) = \left| \sum_{k=1}^N h_k e^{-jk\pi \cos \theta} \right|^2$

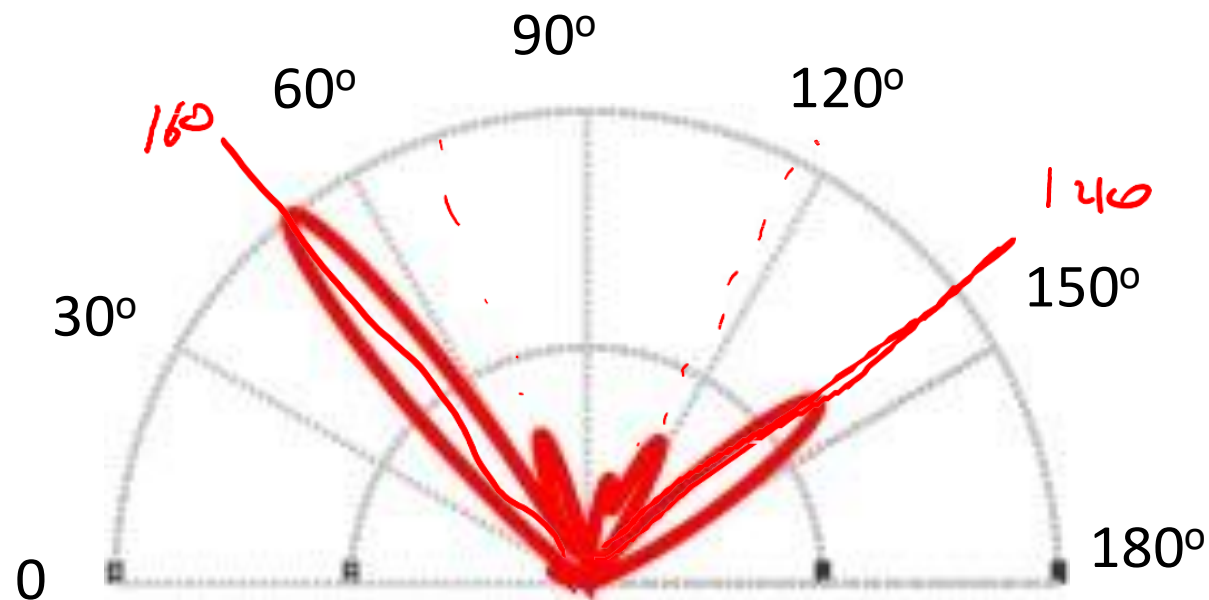
# Method 3: Antenna Arrays (ArrayTrack Paper)

$$\begin{aligned}
 P(\theta) &= \left| \sum_{k=1}^N h_k e^{-jk\pi \cos \theta} \right|^2 & h_k &= \sum_{l=1}^L \alpha_l e^{jk\pi \cos \theta_l - j\phi_l} \\
 P(\theta_1) &= \left| \sum_{k=1}^N h_k e^{-jk\pi \cos \theta_1} \right|^2 \\
 &= \left| \sum_{k=1}^N \left( \sum_{l=1}^L \alpha_l e^{jk\pi \cos \theta_l - j\phi_l} \right) e^{-jk\pi \cos \theta_1} \right|^2 \\
 &= \left| \sum_{k=1}^N \alpha_1 e^{-j\phi_1} + \sum_{k=1}^N \left( \sum_{l=2}^L \alpha_l e^{jk\pi (\cos \theta_l - \cos \theta_1) - j\phi_l} \right) \right|^2 \\
 &= \left| \underbrace{N\alpha_1 e^{-j\phi_1}}_{\downarrow} + \sum_{l=2}^L \alpha_l e^{-j\phi_l} \left( \sum_{k=1}^N e^{jk\pi (\cos \theta_l - \cos \theta_1)} \right) \right|^2 \approx N^2 \alpha_1^2
 \end{aligned}$$

# Method 3: Antenna Arrays (ArrayTrack Paper)

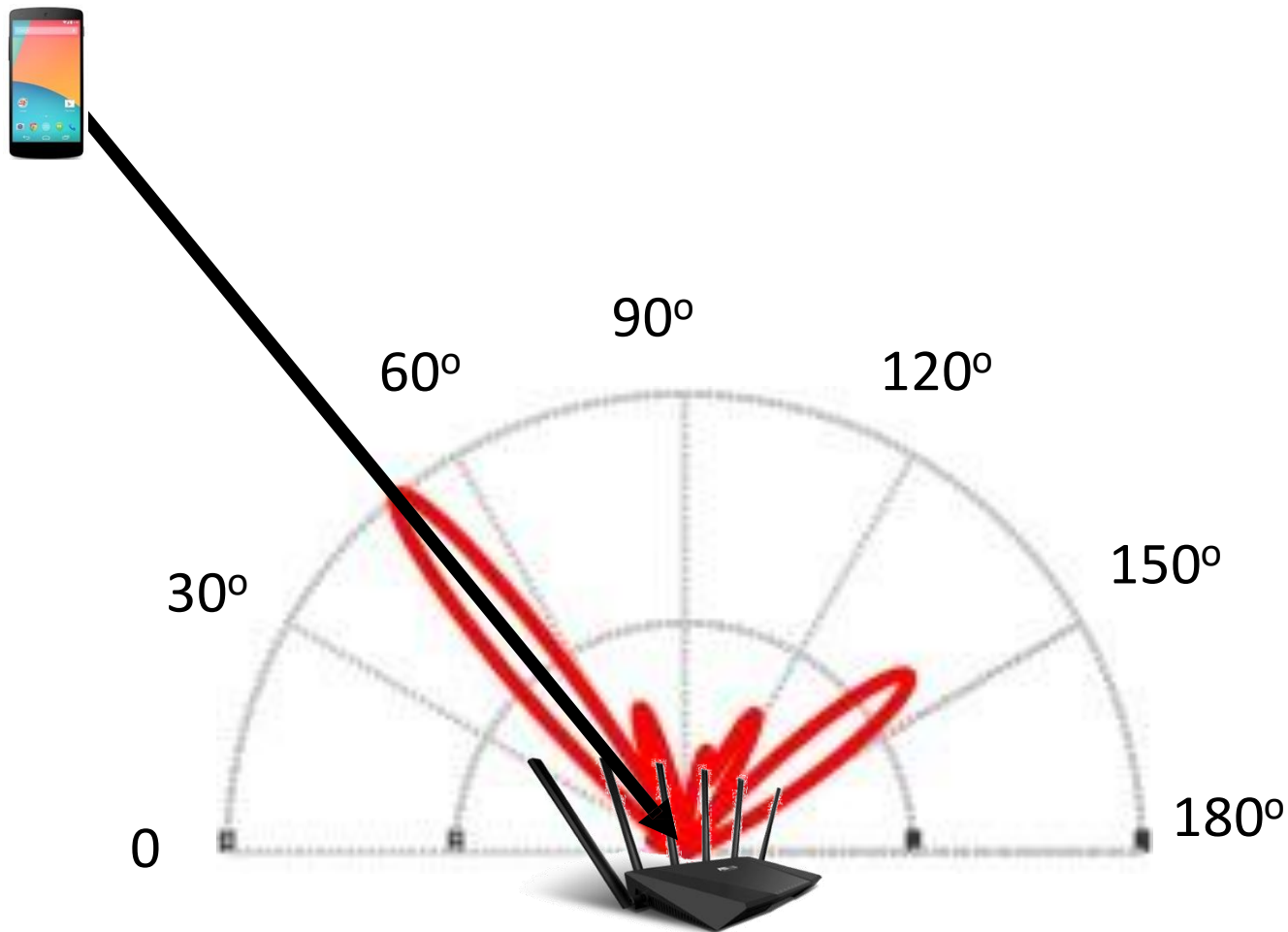
$$h_k = \sum_{l=1}^L \alpha_l e^{jk\pi \cos \theta_l - j\phi_l}$$

$$\text{Multipath Profile: } P(\theta) = \left| \sum_{k=1}^N h_k e^{-jk\pi \cos \theta} \right|^2$$



# Method 3: Antenna Arrays (ArrayTrack Paper)

Which is the Line-of-Sight Path (Direct Path)?  
Strongest Path!



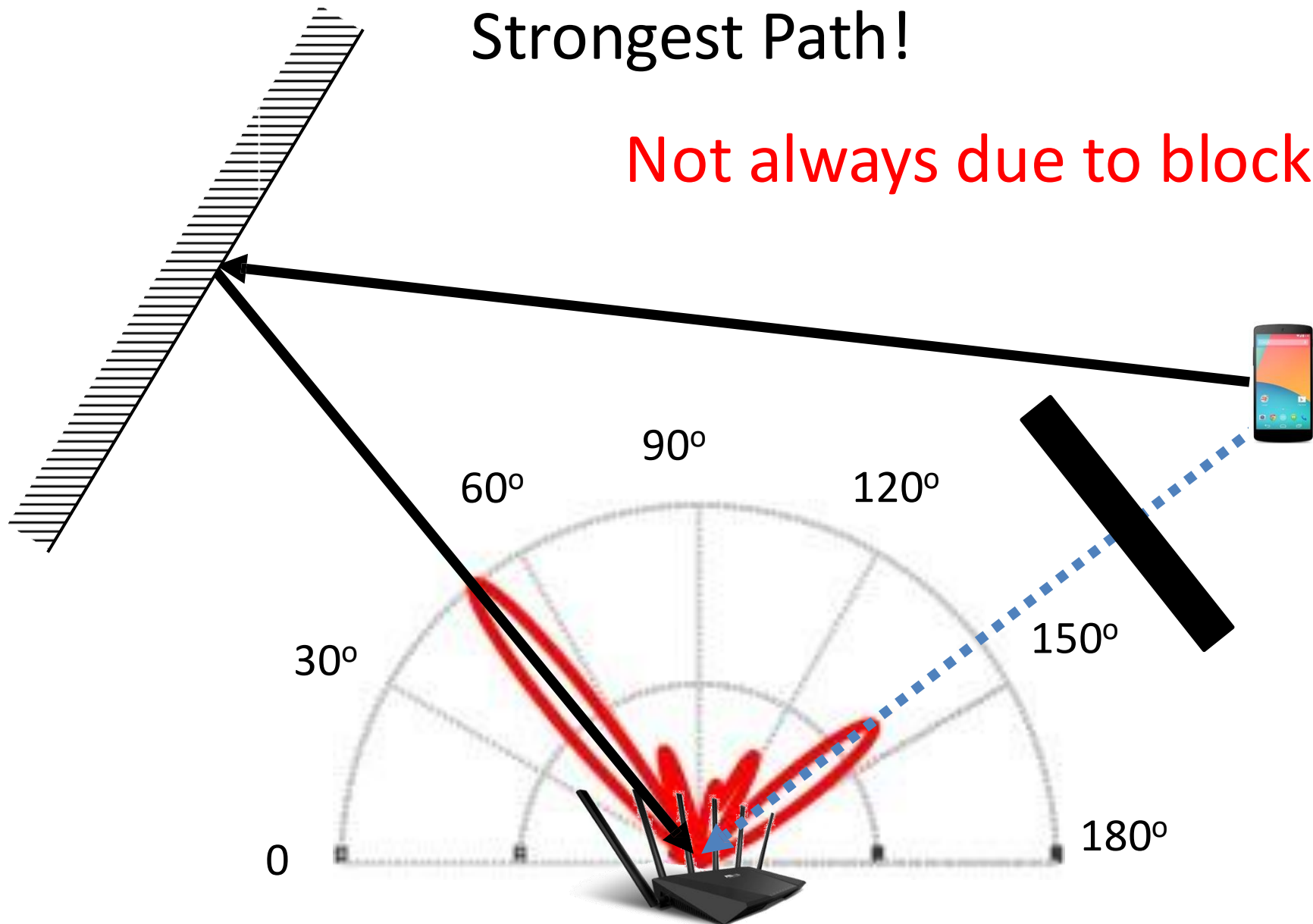


# Method 3: Antenna Arrays (ArrayTrack Paper)

Which is the Line-of-Sight Path (Direct Path)?

Strongest Path!

Not always due to blockage!

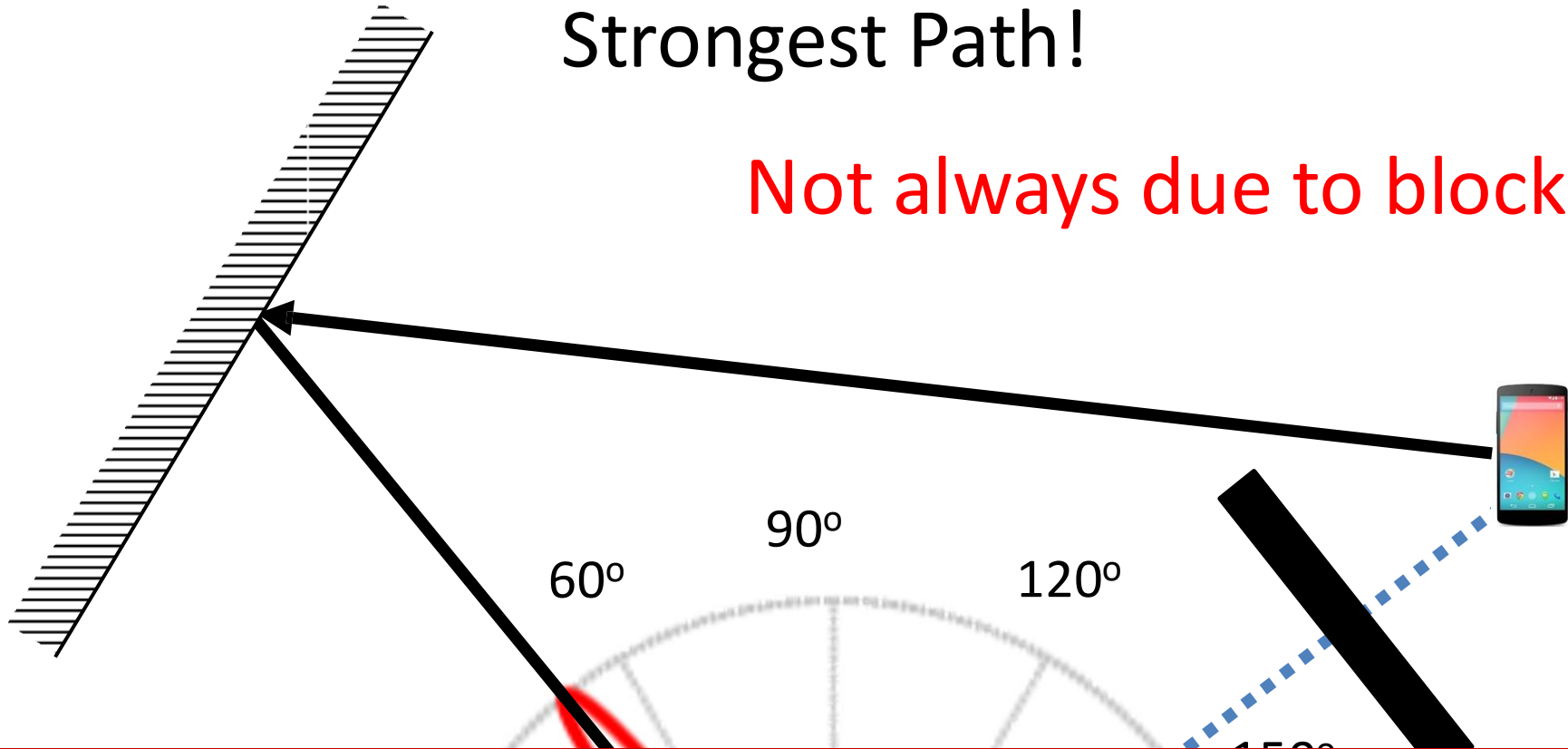


# Method 3: Antenna Arrays (ArrayTrack Paper)

Which is the Line-of-Sight Path (Direct Path)?

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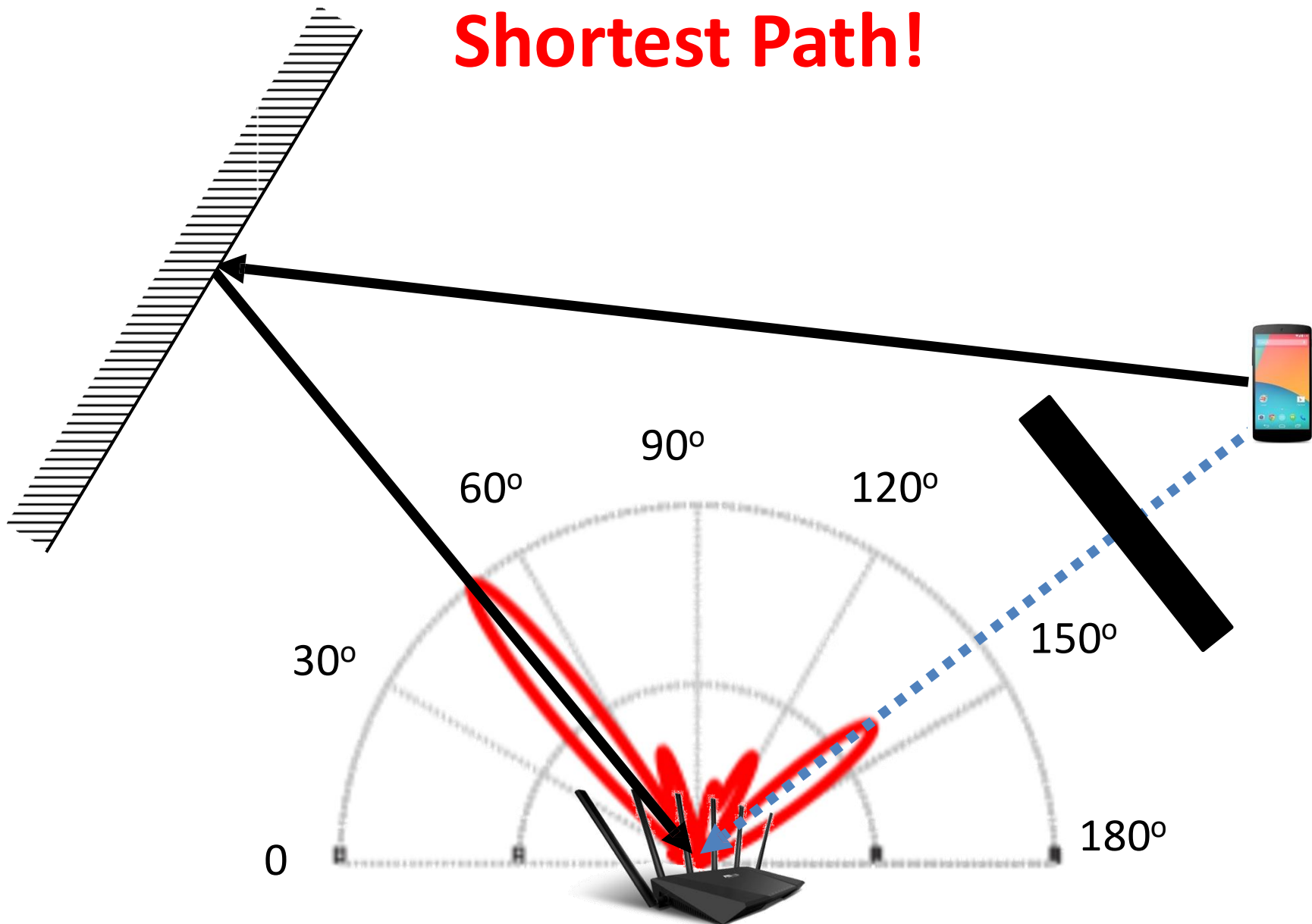
## ArrayTrack: Leverage Mobility

- Line of sight path relatively stable with mobility
- Multipath reflection changes faster with mobility

## Method 3: Antenna Arrays

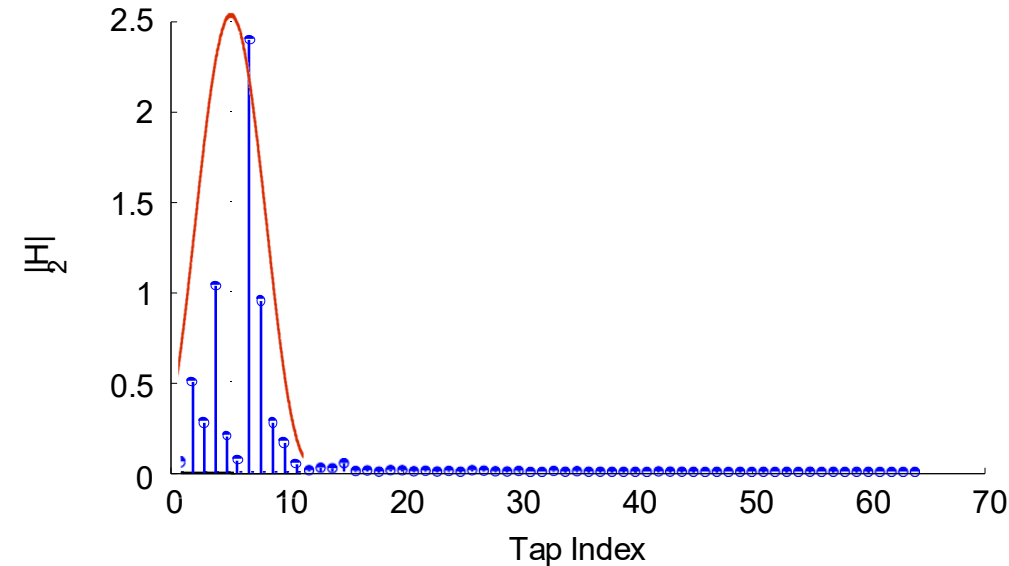
Which is the Line-of-Sight Path (Direct Path)?

**Shortest Path!**



# Method 3: Antenna Arrays

Which is the Shortest Path (Direct Path)?



Multipath Profile vs Time

$$\Delta r = \frac{\Delta d}{c}$$

$$\Delta d = 1m$$

$$c = 3 \times 10^8 m/s$$

$$\Delta r = 3.3ns$$



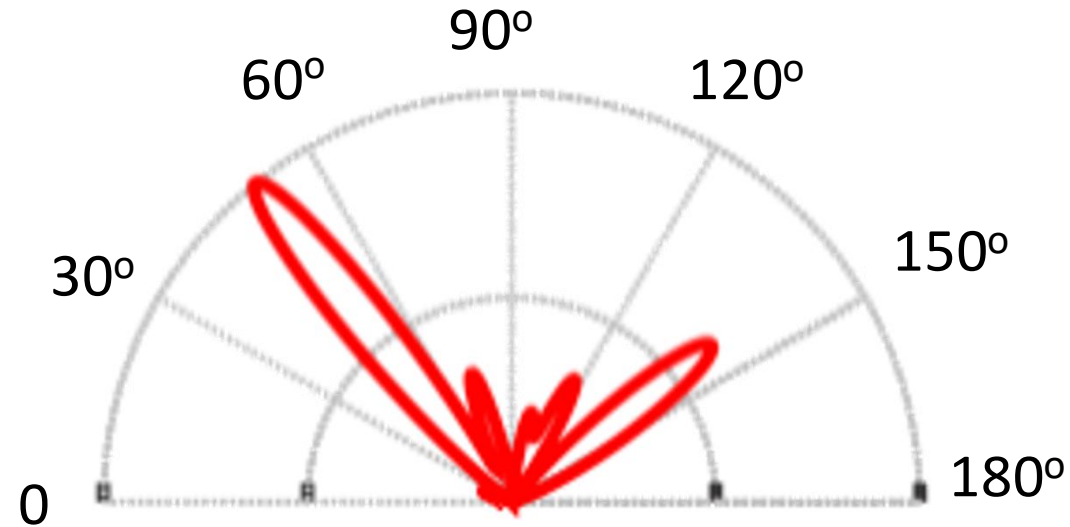
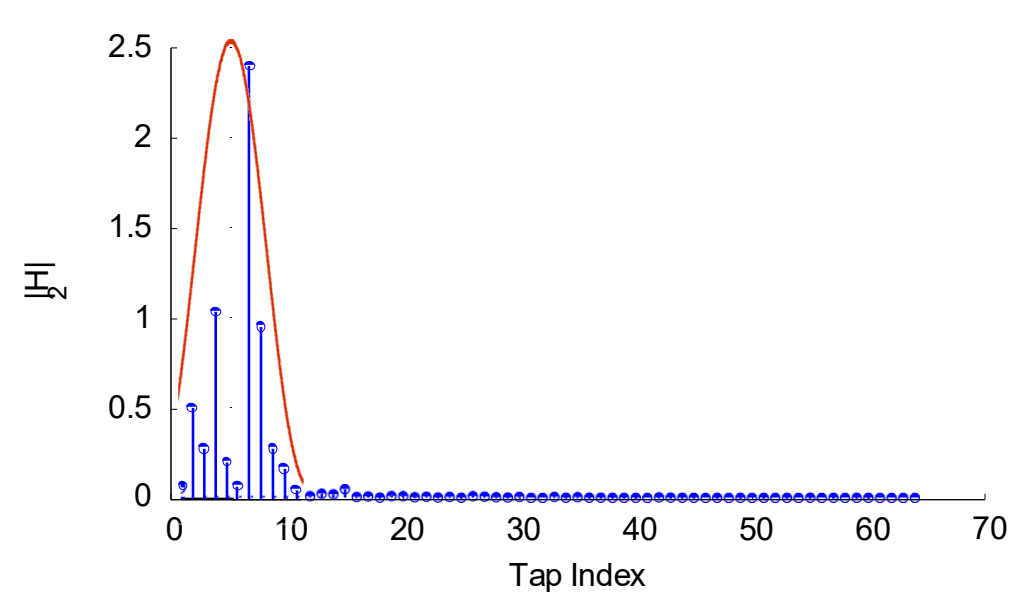
Requires a sampling rate

$$1/\Delta r = 300MHz$$

802.11n bandwidth = 40MHz

# Method 3: Antenna Arrays

Which is the Shortest Path (Direct Path)?



Multipath Profile vs Time

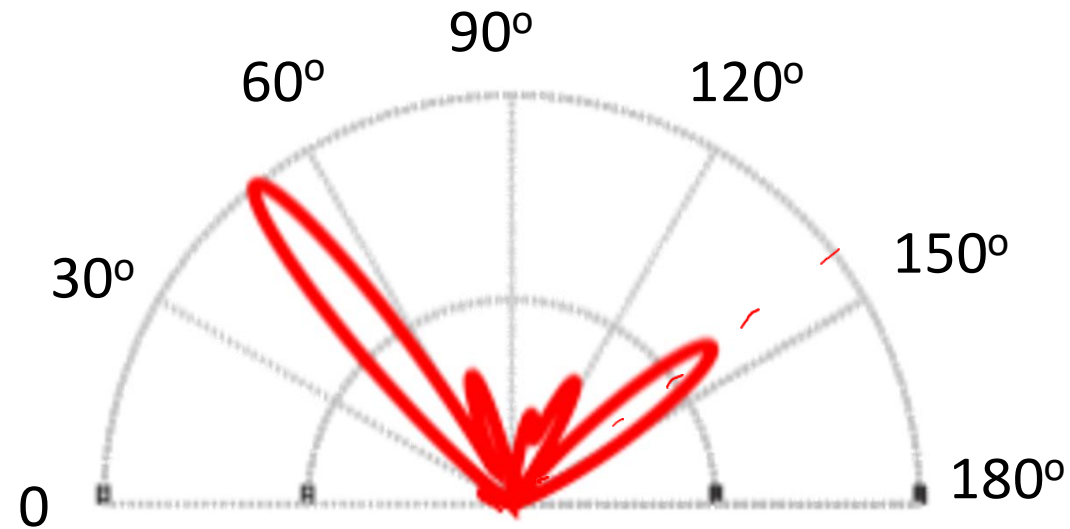
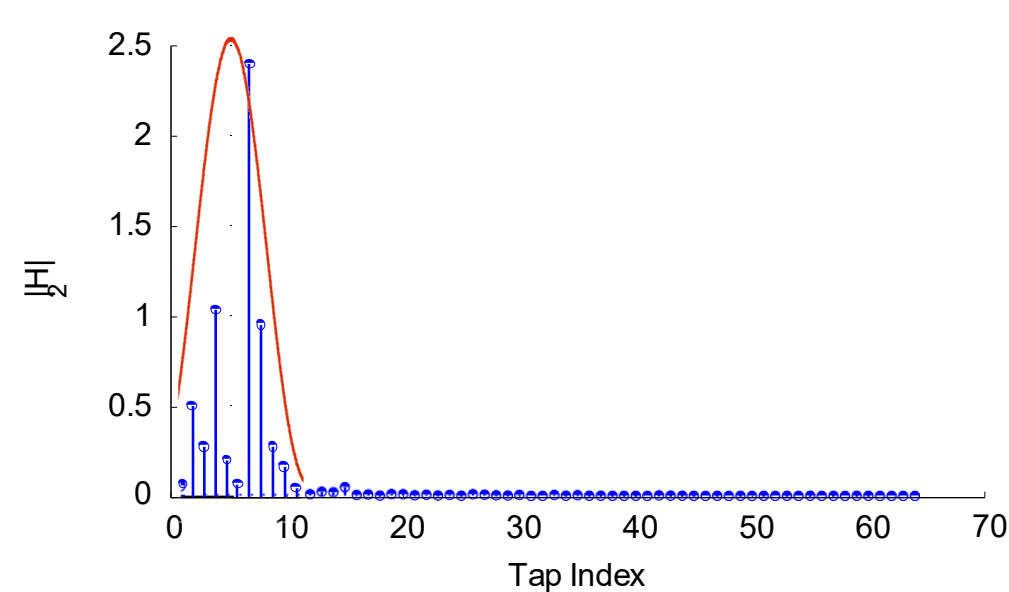
Multipath Profile vs AoA

1. Use multipath profile as a filter to separate different paths
2. Estimate time of arrival of each path
3. Find the shortest path

But How?

# Method 3: Antenna Arrays

Which is the Shortest Path (Direct Path)?



Multipath Profile vs Time

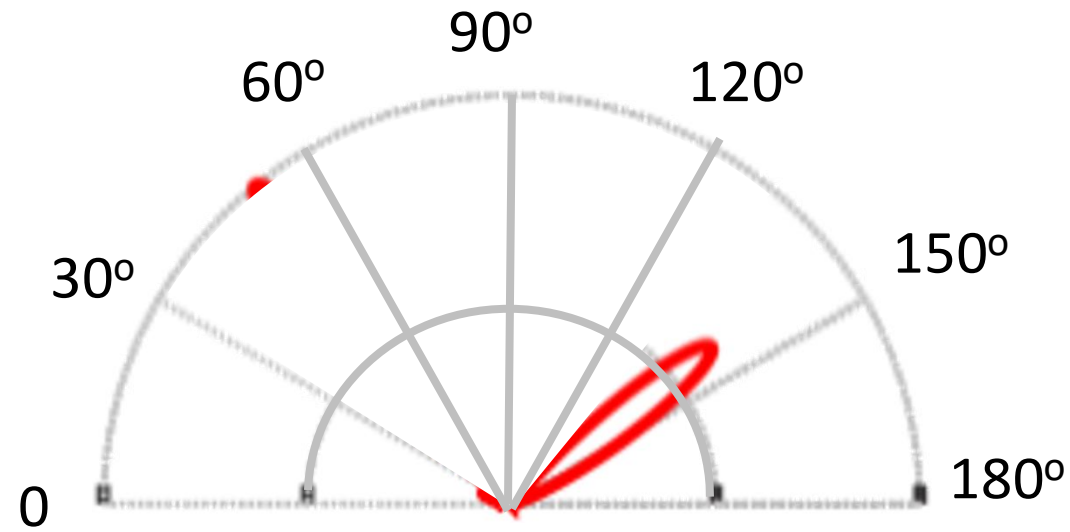
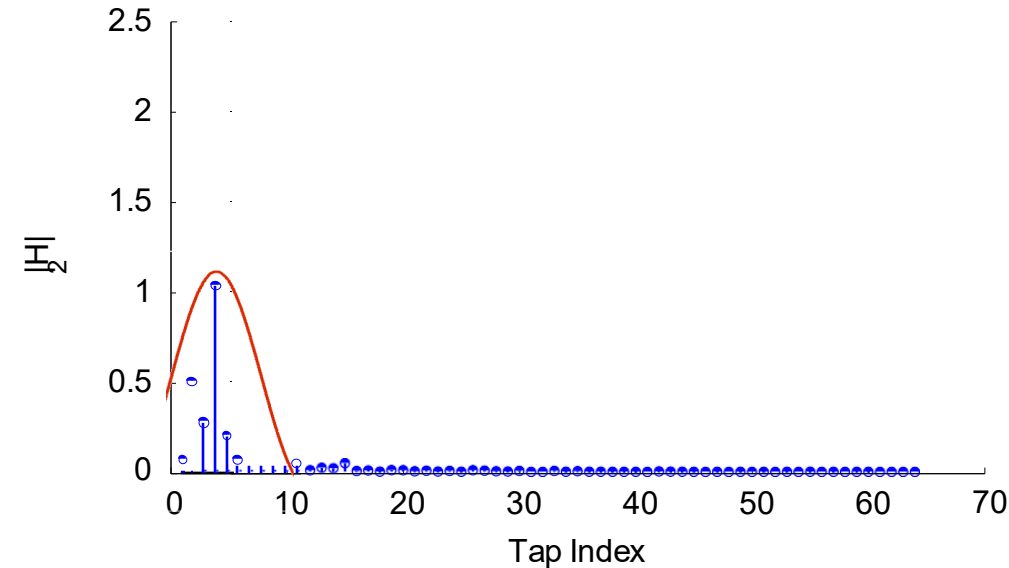
Multipath Profile vs AoA

1. Use multipath profile as a filter to separate different paths

$$y_{\theta_1}(t) = \sum_{k=1}^N y(t) e^{-jk\pi \cos \theta_1}$$

# Method 3: Antenna Arrays

Which is the Shortest Path (Direct Path)?



Multipath Profile vs Time

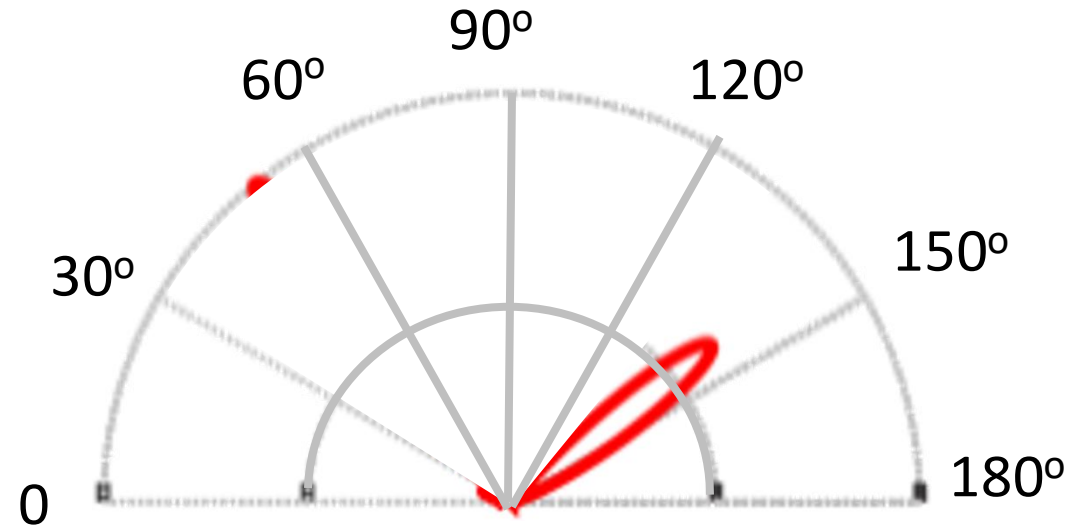
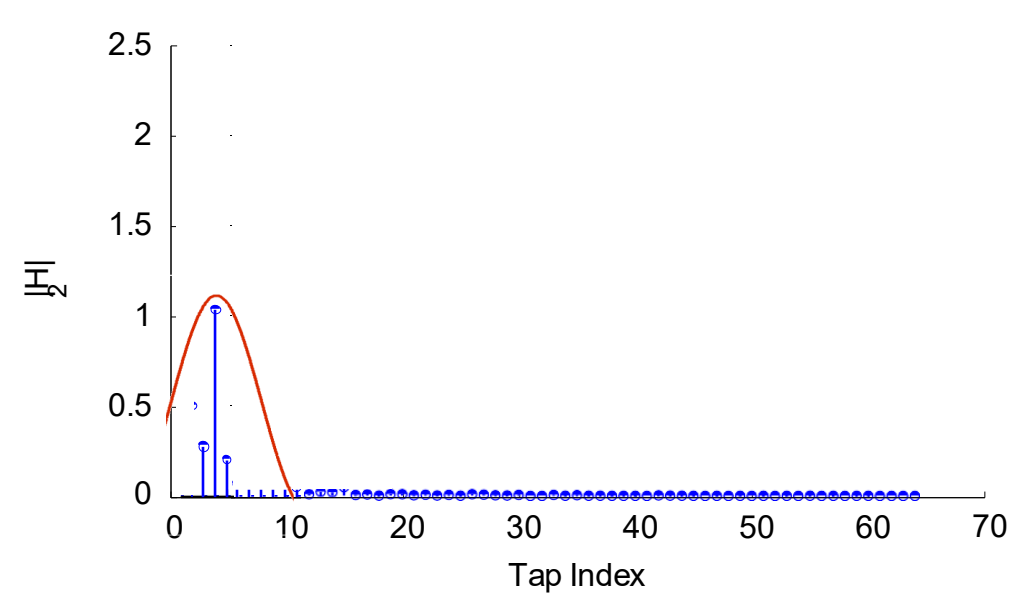
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# Method 3: Antenna Arrays

Which is the Shortest Path (Direct Path)?



Multipath Profile vs Time

Multipath Profile vs AoA

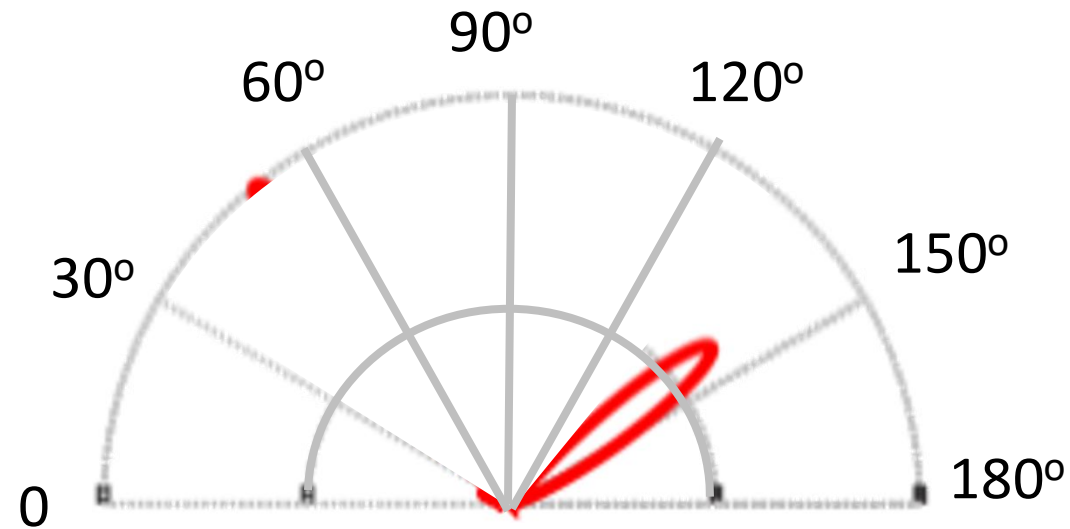
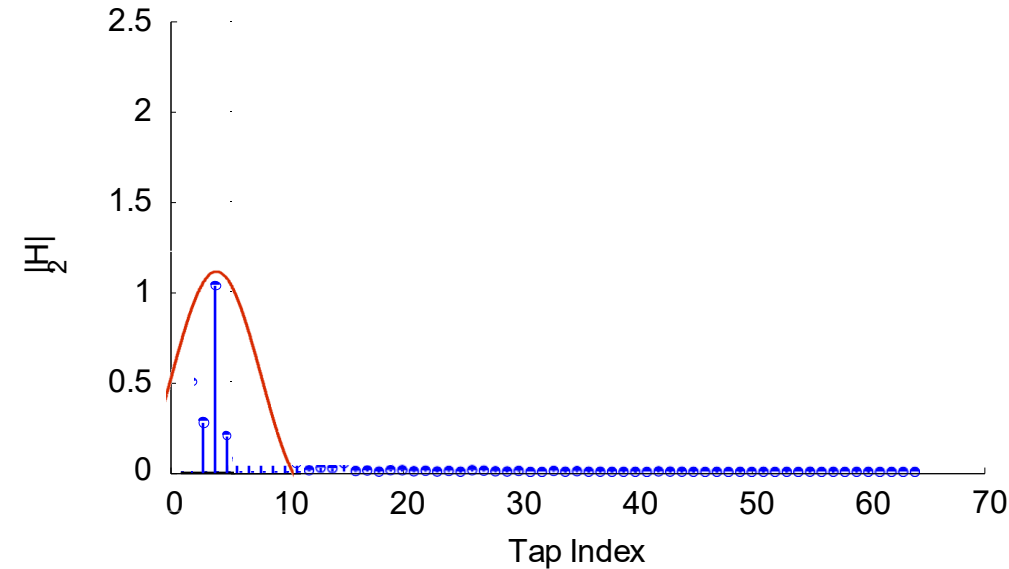
1. Use multipath profile as a filter to separate different paths
2. Estimate time of arrival of each path

Time Resolution still not enough ➡ Use OFDM



# Method 3: Antenna Arrays

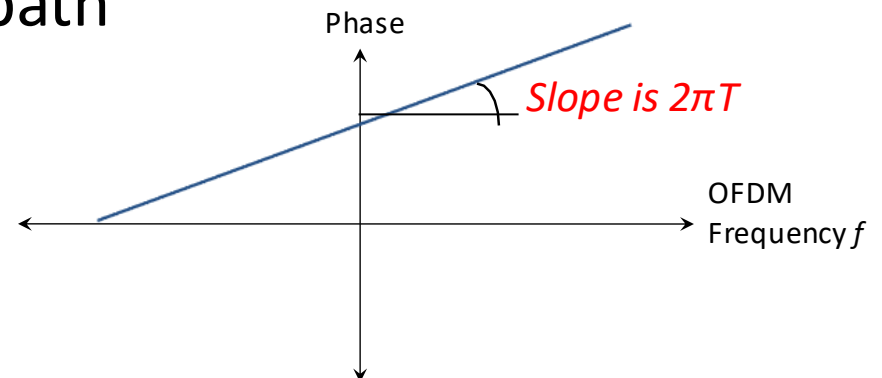
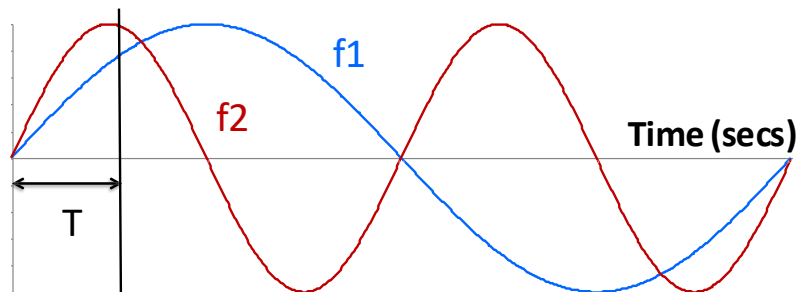
Which is the Shortest Path (Direct Path)?



## Multipath Profile vs Time

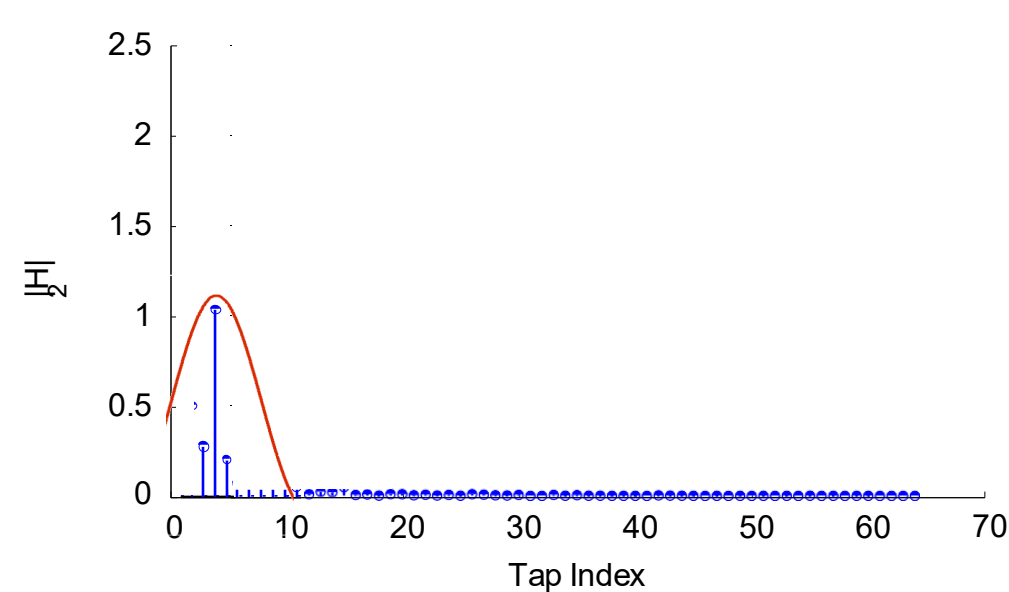
## Multipath Profile vs AoA

1. Use multipath profile as a filter to separate different paths
2. Estimate time of arrival of each path

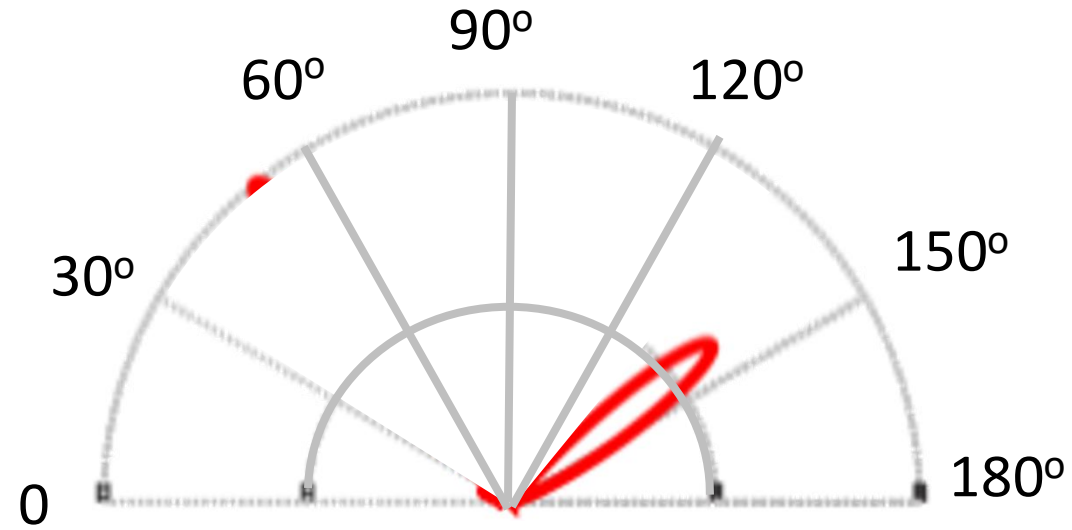


# Method 3: Antenna Arrays

Which is the Shortest Path (Direct Path)?



Multipath Profile vs Time



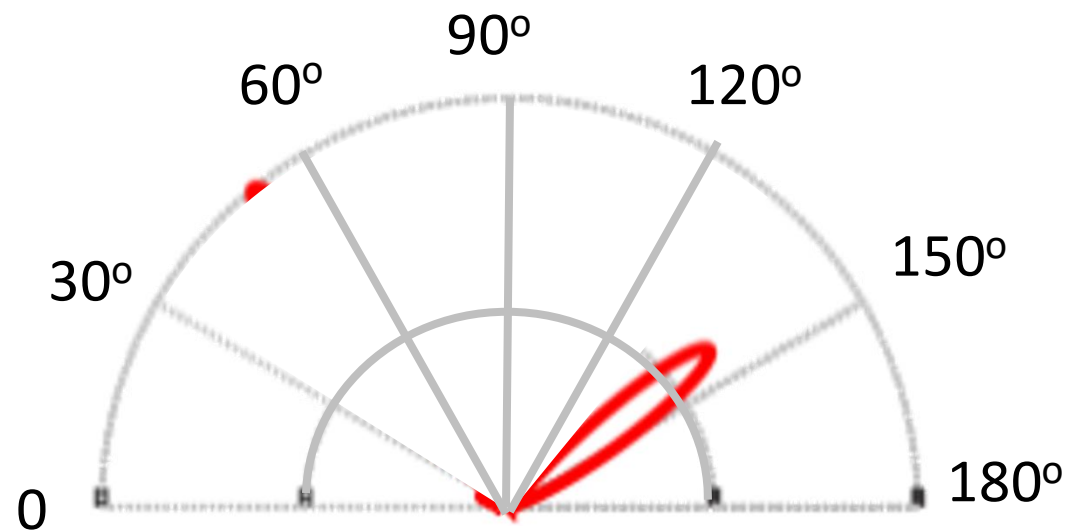
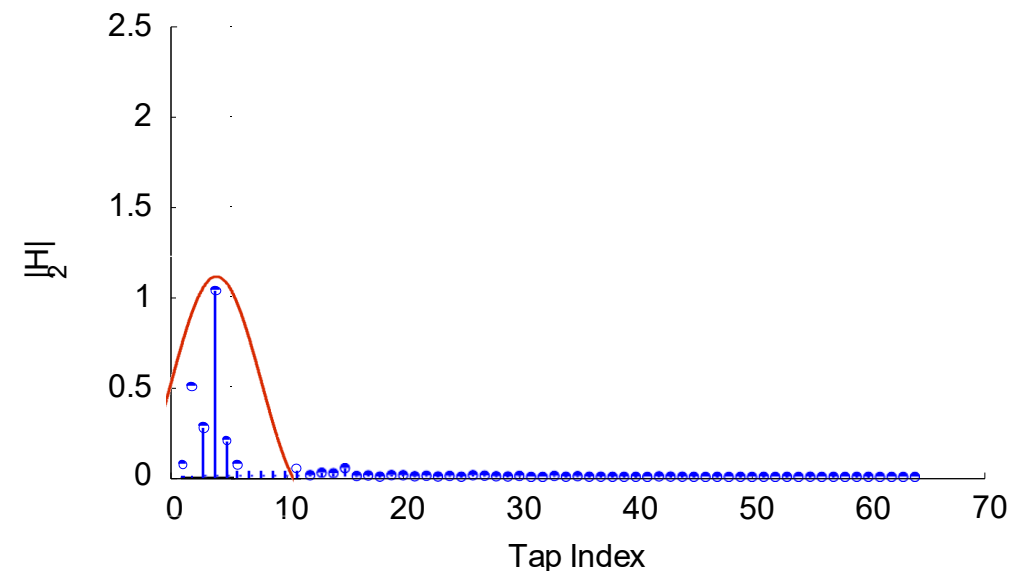
Multipath Profile vs AoA

1. Use multipath profile as a filter to separate different paths
2. Estimate time of arrival of each path
  - Use OFDM to estimate delay from slope of phase vs freq.

But this delay includes packet detection delay & processing delay, not just propagation delay!

# Method 3: Antenna Arrays

Which is the Shortest Path (Direct Path)?



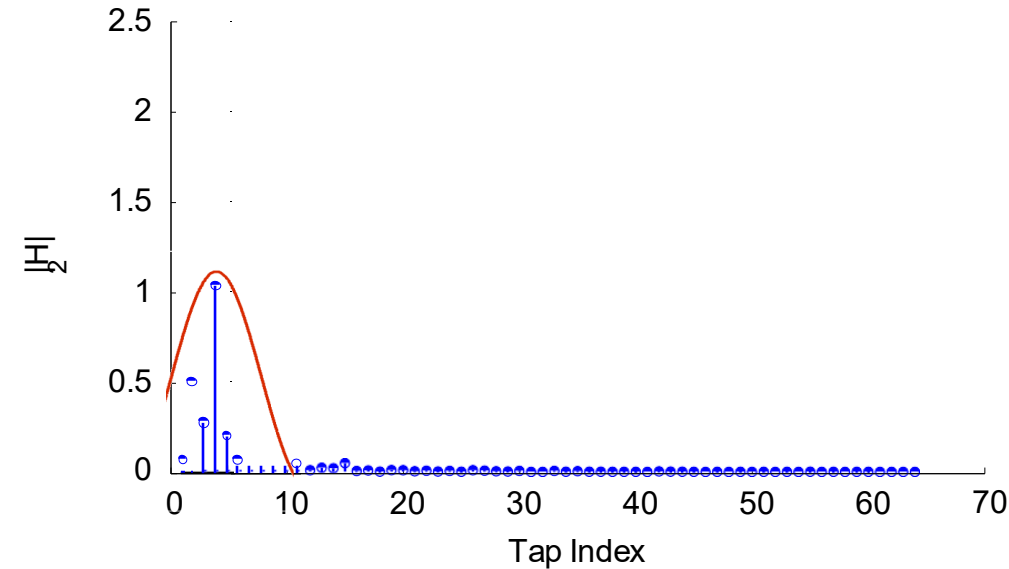
Multipath Profile vs Time

Multipath Profile vs AoA

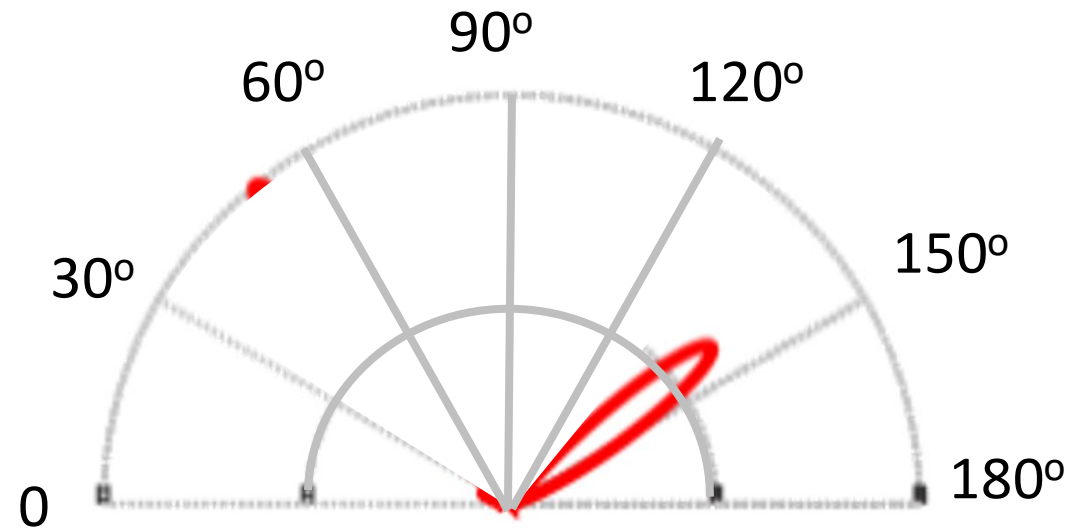
1. Use multipath profile as a filter to separate different paths
2. Estimate time of arrival of each path
  - Use OFDM to estimate delay from slope of phase vs freq.
  - Compute relative delay for different paths!
3. Compare relative delays to find the shortest path

# Method 3: Antenna Arrays

Which is the Shortest Path (Direct Path)?



Multipath Profile vs Time

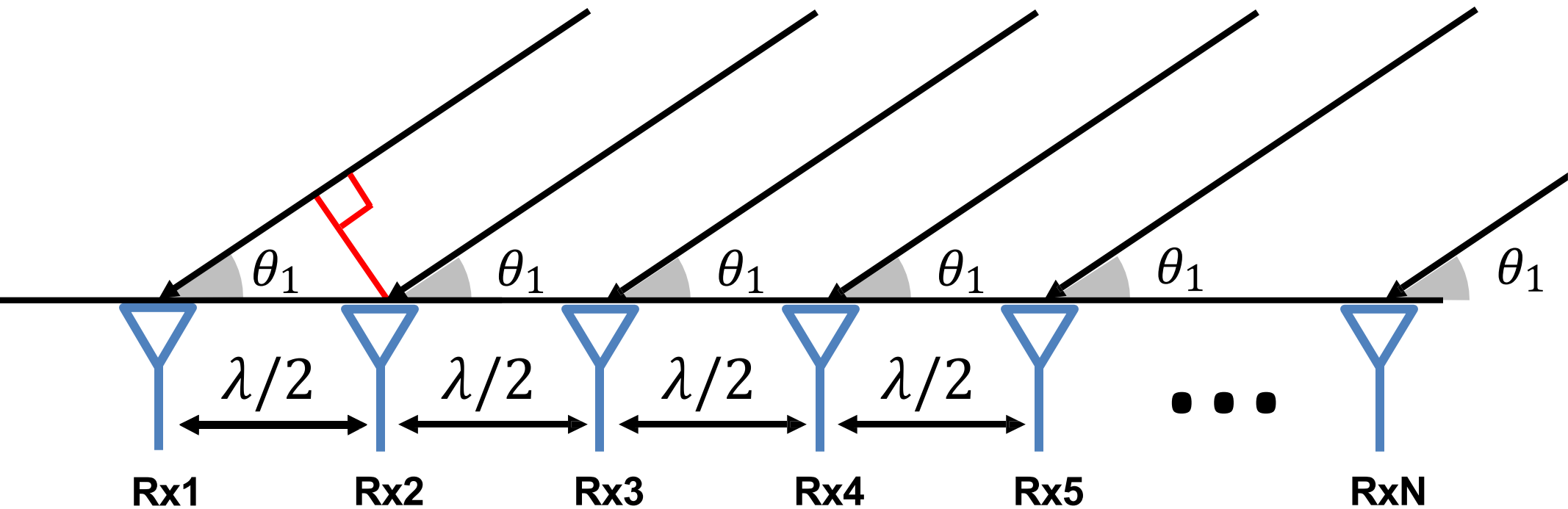


Multipath Profile vs AoA

# Method 3: Antenna Arrays

## AOA-TOF Transform

## Method 3: Antenna Arrays



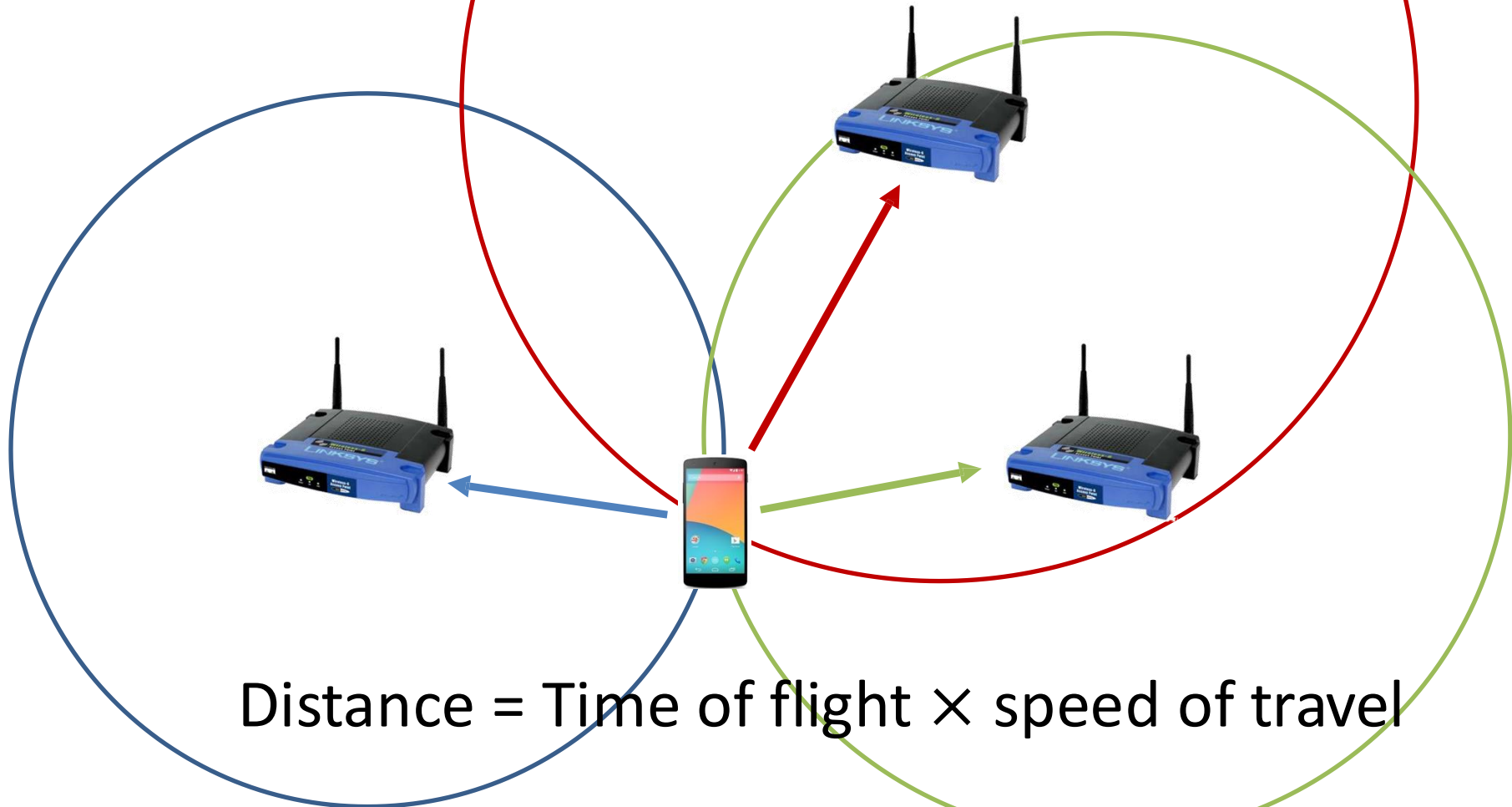
**Pros:** Works with multipath, No need for fingerprinting

**Cons:** Requires more hardware!

Assumes device is sufficiently far such that wavefront is parallel

# Method 4: ToF Based Localization

Measure Time of Flight (ToF) from device to each AP



Measure ToF → Get distance → Trilateration

# Method 4: ToF Based Localization

Measure Time of Flight (ToF) from device to each AP

## Challenges:

- How do you know when signal was transmitted?



Measure Round Trip

- How about packet detection delay & processing delay?
  - Use OFDM to correct for packet detection delay
  - Estimate and calibrate for processing delay

**Not Practical!**



# Method 4: ToF Based Localization

Measure Time of Flight (ToF) from device to each AP

## Challenges:

- Resolution limited by sampling rate (bandwidth)!

$$\Delta d = \Delta \tau \times c$$

$$802.11n \text{ bandwidth} = 40MHz \Rightarrow \Delta \tau = 25ns \Rightarrow \Delta d = 12.5 m$$

- Other systems than WiFi can get accurate ToF:
  - UWB: Ultra-Wide Band
  - FMCW: Frequency Modulated Carrier Wave

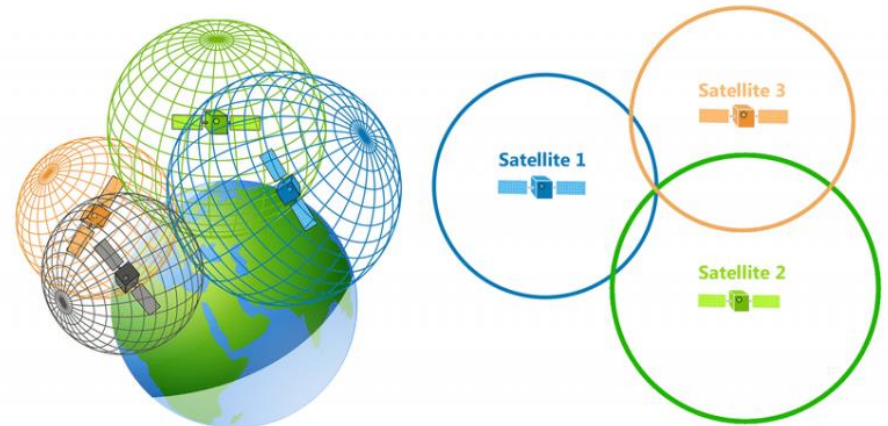
**Not Supported in WiFi**  
**(Will discuss in future lectures)**

# Method 4: ToF Based Localization

Measure Time of Flight (ToF) from device to each AP

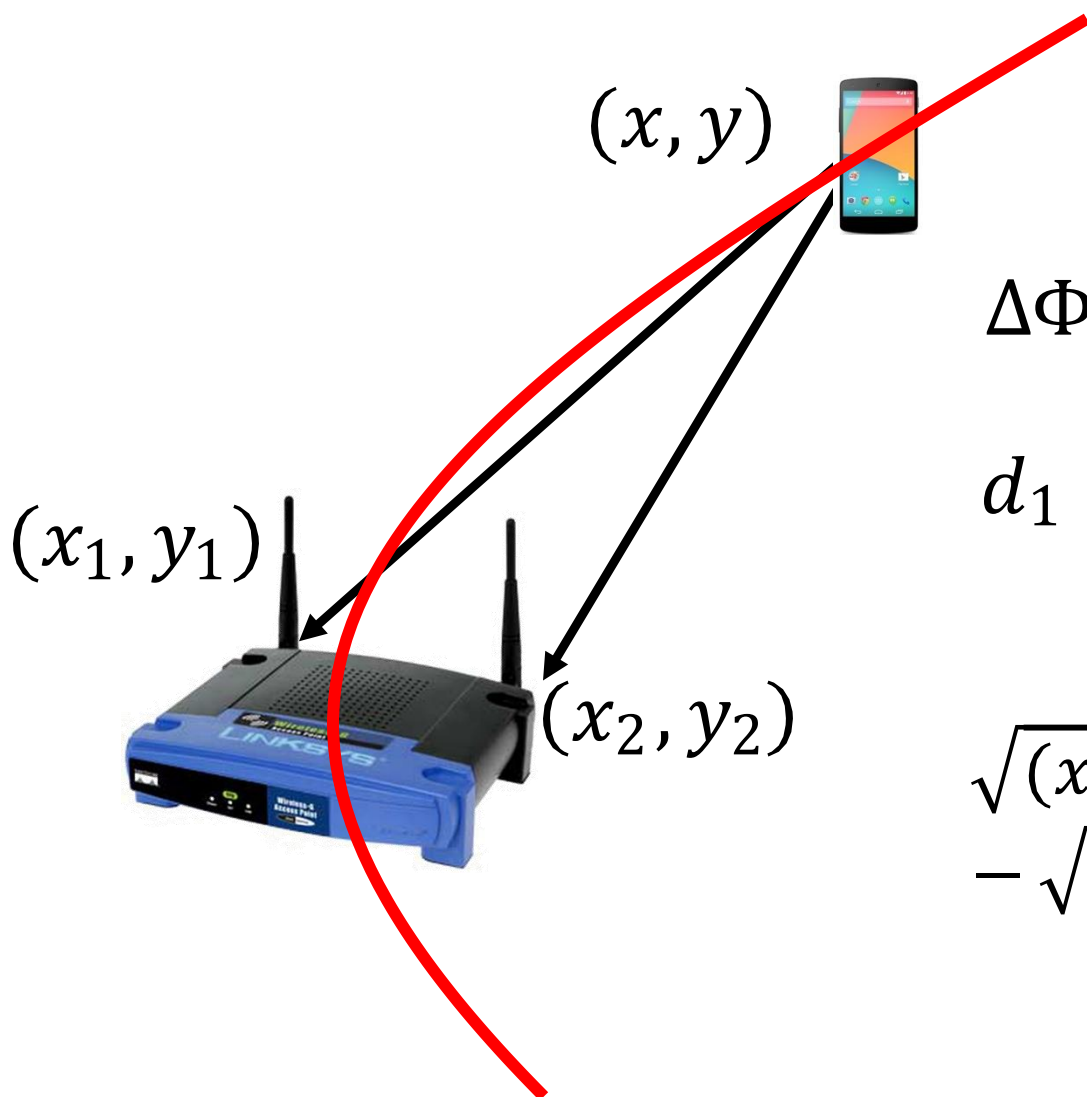
## Challenges:

- How do you know when signal was transmitted?
- How GNSS Positioning work then?
- Satellite beacon sends its positions and time sent.
- Device estimates time-difference between satellite clock and device clock → Distance
- All satellites are perfectly synchronized
- Four unknown –  $x, y, z$ , clock bias → 4 equations from at least 4 satellites.



# Method 5: TDoA Based Localization

Measure Time Difference of Arrival (TDoA) from device to AP's antennas



$$h_1 \propto e^{-j2\pi\frac{d_1}{\lambda}} \quad h_2 \propto e^{-j2\pi\frac{d_2}{\lambda}}$$

$$\Delta\Phi = \angle h_2 - \angle h_1$$

$$d_1 - d_2 = \lambda\Delta\Phi/2\pi = d'$$

**Equation of Hyperbola**

$$\sqrt{(x - x_1)^2 + (y - y_1)^2} - \sqrt{(x - x_2)^2 + (y - y_2)^2} = d'$$

# Method 5: TDoA Based Localization

Measure Time Difference of Arrival (TDoA) from device to AP's antennas

